The inverse functions $\sin^{-1} x$, $\cos^{-1} x$, $\tan^{-1} x$, $\text{cosec}^{-1}(x)$, $\sec^{-1}(x)$, $\cot^{-1}(x)$ are called inverse circular functions. For the function $y = \sin x$, there are infinitely many angles x which satisfy $\sin x = t$, $-1 \leq t \leq 1$. Of these infinite set of values, there is one which lies in the interval $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$. This angle is called the **principal angle** and denoted by $\sin^{-1} t$. The principal value of an inverse function is that value of the general value which is numerically least. It may be positive or negative. When there are two values, one is positive and the other is negative such that they are numerically equal, then the principal value is the positive one.

We shall illustrate below the restricted domains, ranges of trigonometric functions and the domains, ranges of the corresponding inverse functions.

(i)	$\sin x : \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \to [-1, 1]$;	$\sin^{-1} x : [-1, 1] \to \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
(ii)	$\cos x : [0, \pi] \to [-1, 1]$;	$\cos^{-1} x : [-1, 1] \to [0, \pi]$
(iii)	$\tan x : \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \to (-\infty, \infty)$;	$\tan^{-1} x : (-\infty, \infty) \to \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
(iv)	$\cot x : (0, \pi) \to (-\infty, \infty)$;	$\cot^{-1} x : (-\infty, \infty) \to (0, \pi)$
(v)	$\text{cosec} x : \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] - \{0\} \to \mathbb{R} - (-1, 1)$;	$\text{cosec}^{-1} x : \mathbb{R} - (-1, 1) \to \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] - \{0\}$
(vi)	$\sec x : [0, \pi] - \left\{\frac{\pi}{2}\right\} \to \mathbb{R} - (-1, 1)$;	$\sec^{-1} x : \mathbb{R} - (-1, 1) \to [0, \pi] - \left\{\frac{\pi}{2}\right\}$

(i) $\sin^{-1} x$ does not mean $\frac{1}{\sin x}$.

(ii) Another notation for $\sin^{-1} x$ is $\arcsin x$ due to **Sir. John FW Herschel (1813)**.

(iii) While discussing the inverse of sine function, we confined to $y = \sin x, -\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$ and
$x = \sin^{-1} y, -1 \leq y \leq 1$

(iv) The graph of inverse function f^{-1} is the reflexion of the graph of f about the line $y = x$. Thus, if $(a, b) \in f$ then $(b, a) \in f^{-1}$.

Principal values of inverse trigonometric functions are listed below:

Principal value for $x \geq 0$	Principal value for $x < 0$
$0 \leq \sin^{-1}(x) \leq \frac{\pi}{2}$	$-\frac{\pi}{2} \leq \sin^{-1}(x) < 0$
$0 \leq \cos^{-1}(x) \leq \frac{\pi}{2}$	$\frac{\pi}{2} < \cos^{-1}(x) \leq \pi$
$0 \leq \tan^{-1}(x) < \frac{\pi}{2}$	$-\frac{\pi}{2} < \tan^{-1}(x) < 0$
$0 < \cot^{-1}(x) \leq \frac{\pi}{2}$	$-\frac{\pi}{2} < \cot^{-1}(x) < 0$
$0 \leq \sec^{-1}(x) < \frac{\pi}{2}$	$\frac{\pi}{2} < \sec^{-1}(x) \leq \pi$
$0 < \text{cosec}^{-1}(x) \leq \frac{\pi}{2}$	$-\frac{\pi}{2} < \text{cosec}^{-1}(x) < 0$